

NORTHVALE THERAPEUTICS | MEDICAL AFFAIRS

Obesity: Disease Burden, Pathophysiology and Management

An evidence overview for internal scientific exchange

Document DK-ARC-2600002 · Version 0.1 · September 2026

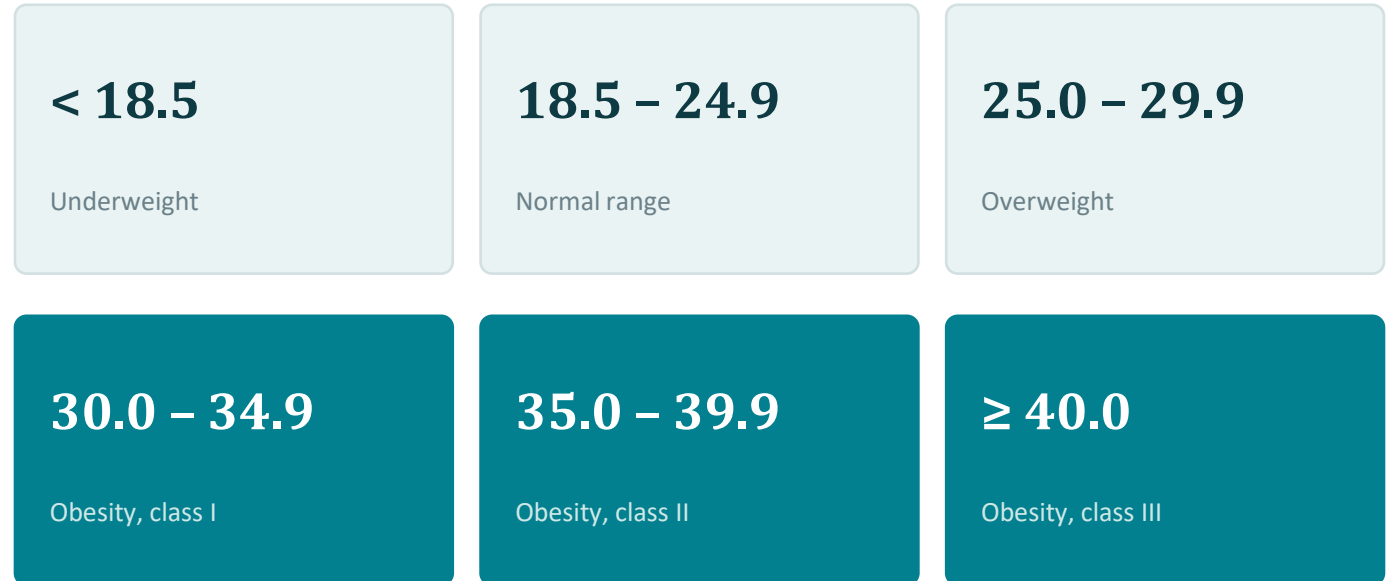
Fictional test material. All data, products and citations are invented and are not for distribution or clinical use.



Defining and classifying obesity

Obesity is characterised by excess adipose accumulation sufficient to impair health [1]. Body mass index remains the primary population screening metric, calculated as weight in kilograms divided by height in metres squared.

BMI does not distinguish adipose tissue from lean mass, and performs inconsistently across ancestry groups and at the extremes of stature [1,2]. Waist circumference and waist-to-height ratio are recommended as adjunct measures [2].

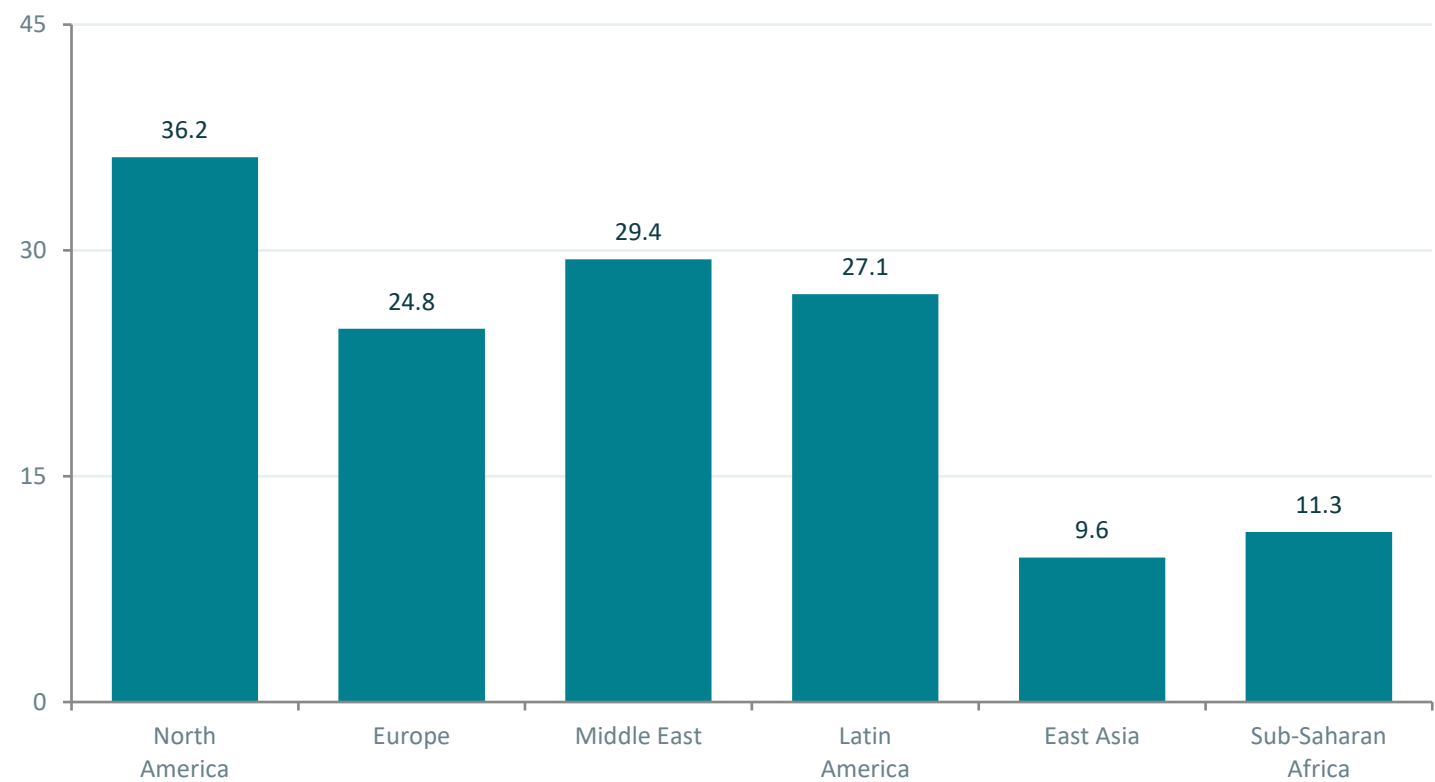


BMI, kg/m^2

References

1. Halvorsen RT, Ibarra-Núñez M, Sandoval KP. Redefining adiposity thresholds in adult screening. *J Metab Med*. 2021;48(3):215–229. doi:10.4821/jmm.2021.0348
2. Okonkwo BA, Lindqvist H. Limitations of body mass index as a compositional proxy. *Ann Obes Sci*. 2020;12(7):1104–1117. doi:10.5507/aos.2020.1104

Prevalence is rising across every region



Key observations

- Age-standardised prevalence has approximately doubled in adults since 1990 across most reporting regions [3].
- Modelling projects 1.9 billion adults above the class I threshold by 2035 under current trajectories [4].
- Urban incidence exceeds rural incidence in high-income settings, with the gradient reversing in several low-income settings [5].
- Regional figures conceal substantial within-country variation by income decile [3,5].

References

3. Duarte FM, Reinholt AS, Chaudhary V. Regional prevalence trends in adult obesity, 2005–2022. *Glob Health Policy Rev.* 2023;31(2):88–103. doi:10.1188/ghpr.2023.0088

4. Mbeki NT, Salonen EJ. Projected adiposity burden to 2035: a modelling analysis. *Int Rev Endocrinol.* 2023;55(4):402–419. doi:10.3310/ire.2023.0402

5. Vasquez-Lund P, Achterberg J, Nakamura Y. Urban–rural divergence in metabolic disease incidence. *J Popul Metab Health.* 2022;9(1):33–47. doi:10.2217/jpmh.2022.0033

Energy homeostasis and adipose expansion

01

Energy surplus

Sustained positive energy balance drives triglyceride storage in subcutaneous depots [6].

02

Hypothalamic signalling

Leptin and insulin afferent signalling attenuates, raising the defended body-weight set point [6].

03

Adipocyte hypertrophy

Storage capacity is exceeded; lipid deposits ectopically in liver and skeletal muscle [7].

04

Metabolic dysregulation

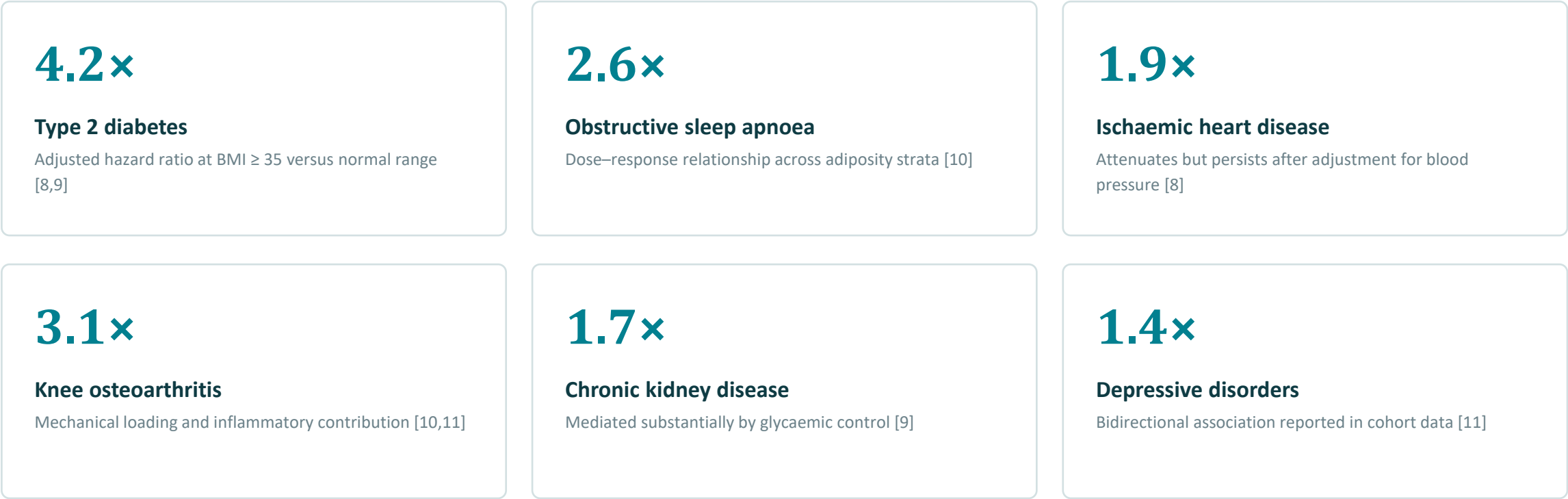
Local inflammation and insulin resistance establish a self-reinforcing cycle [7,2].

The defended set point explains why weight regain follows energy restriction in the majority of participants [6].

References

2. Okonkwo BA, Lindqvist H. Limitations of body mass index as a compositional proxy. *Ann Obes Sci.* 2020;12(7):1104–1117. doi:10.5507/aos.2020.1104
6. Farrell KO, Bertrand LM. Hypothalamic regulation of energy homeostasis: a contemporary review. *Int Rev Endocrinol.* 2021;53(6):611–634. doi:10.3310/ire.2021.0611
7. Sørensen AK, Villanueva RC, Teixeira PDS. Adipose tissue expansion and ectopic lipid deposition. *Cell Metab Rep.* 2022;18(2):140–159. doi:10.7012/cmr.2022.0140

Associated conditions and relative risk



Relative risks are illustrative and drawn from pooled cohort estimates; confidence intervals are omitted for clarity.

References

8. Ekelund HJ, Moreau CL. Cardiometabolic risk stratification in class II and III obesity. Eur J Cardiometaab Health. 2022;27(9):1201–1218. doi:10.4419/ejcmh.2022.1201
9. Brantley DW, Nishimura A, Okafor CE. Incident type 2 diabetes across adiposity strata: a pooled cohort analysis. J Metab Med. 2023;50(1):19–36. doi:10.4821/jmm.2023.0019
10. Castellanos IR, Petrov YA. Obstructive sleep apnoea and adiposity: dose–response evidence. Respir Metab J. 2021;14(5):388–401. doi:10.6644/rmj.2021.0388
11. Whitcombe LE, Adeyemi TO. Osteoarthritis progression and mechanical loading in elevated BMI. Clin Musculoskelet Rev. 2020;38(3):277–290. doi:10.9021/cmr.2020.0277

Direct and indirect cost burden

\$173bn

Annual direct medical cost

Attributable spend across inpatient, outpatient and pharmacy settings [12]

2.4%

Share of national health expenditure

Median across twelve high-income reporting systems [12]

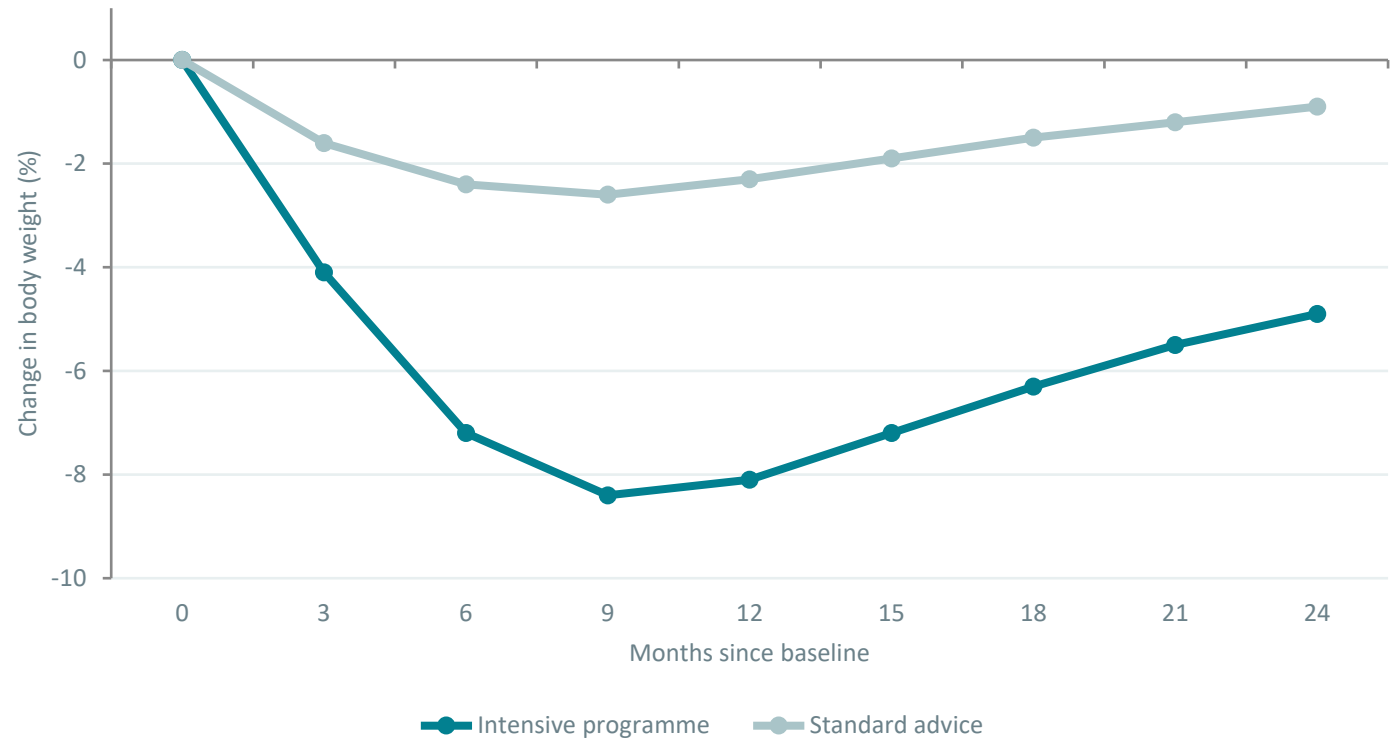
\$89bn

Indirect productivity loss

Absenteeism, presenteeism and premature workforce exit [13]

Cost attribution methodology varies materially between published estimates. Prevalence-based approaches and incidence-based approaches produce figures that differ by a factor of two or more for the same population, and cross-study comparison is therefore inadvisable without harmonisation [12,13].

Structured lifestyle intervention at 24 months



Interpretation

- Nadir weight loss occurs at month nine in the intensive arm, with partial regain thereafter [14].
- Approximately 60% of maximal loss is retained at 24 months among programme completers [14,15].
- Attrition reaches 34% by month 18 and is the principal limitation of the evidence base [15].
- Glycaemic benefit persists beyond the weight trajectory in participants with prediabetes [9].

Mechanistic classes in weight management

Incretin pathway

- Mean reduction 12–18% at 68 weeks [16]
- Gastrointestinal events most common [17]
- Discontinuation 7–14% in trial settings [17]

Central appetite modulation

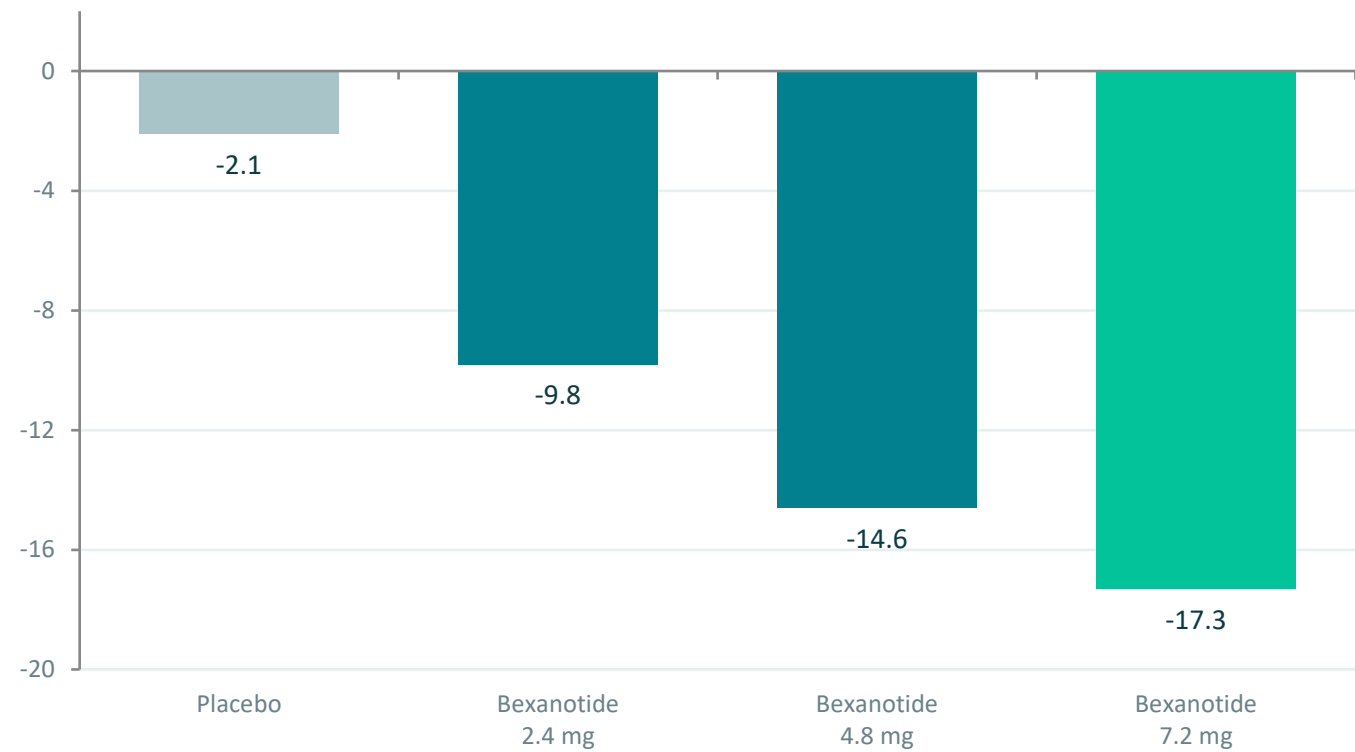
- Mean reduction 5–9% at 52 weeks [16]
- Neuropsychiatric monitoring required [17]
- Contraindicated in uncontrolled hypertension [16]

Absorption inhibition

- Mean reduction 3–5% at 52 weeks [16]
- Steatorrhoea limits adherence [18]
- Fat-soluble vitamin supplementation advised [18]

Adherence at 12 months remains below 45% across all three classes in real-world cohorts, and is the dominant determinant of realised benefit [18].

ZENITH-3: primary endpoint at week 68



Trial design

- Randomised, double-blind, placebo-controlled, 68 weeks; n = 2,140 across 94 sites [19].
- Adults with BMI ≥ 30 , or ≥ 27 with at least one weight-related comorbidity [19].
- All active arms met the primary endpoint versus placebo ($p < 0.001$) [20].
- Dose–response was preserved across prespecified subgroups [20].

Bexanotide is a fictional investigational compound. These data are invented for testing purposes.

Guidance and practice implications

1

Stage, do not just measure

Classification should combine anthropometry with functional and metabolic staging rather than BMI alone [21].

2

Escalate on response

Treatment intensity should follow documented response at defined review points rather than baseline BMI [21,22].

3

Plan for chronicity

Weight regain after intervention withdrawal is the expected course; follow-up should extend beyond 24 months [23].

4

Document the pathway

Local care pathways should record the rationale for each escalation step to support later audit [24].

Sources cited in sections 05–09

9. Brantley DW, Nishimura A, Okafor CE. Incident type 2 diabetes across adiposity strata: a pooled cohort analysis. *J Metab Med*. 2023;50(1):19–36. doi:10.4821/jmm.2023.0019
12. Rautenbach SJ, Delgado-Ferreira N. Direct medical costs attributable to adiposity-related disease. *Glob Health Policy Rev*. 2022;30(6):512–530. doi:10.1188/ghpr.2022.0512
13. Kaur NS, Lindholm PE, Escobar TJ. Indirect productivity losses in metabolic disease: a multi-country estimate. *Health Econ Perspect*. 2023;41(2):205–223. doi:10.3355/hep.2023.0205
14. Nyberg AC, Rasheed FI. Structured lifestyle intervention at 24 months: pooled outcomes. *Clin Nutr Rep*. 2022;19(4):455–472. doi:10.8180/cnr.2022.0455
15. Thorvaldsen EM, Guo L, Amorim RB. Weight regain trajectories following behavioural programmes. *Ann Obes Sci*. 2023;15(2):190–208. doi:10.5507/aos.2023.0190
16. Panayiotou SK, Reddick JM. Mechanistic classes in contemporary weight management pharmacotherapy. *J Metab Med*. 2023;50(5):601–625. doi:10.4821/jmm.2023.0601
17. Isaksson MR, Oyelaran BT. Tolerability profiles across incretin-pathway agents. *Eur J Cardiometa Health*. 2023;28(3):310–329. doi:10.4419/ejcmh.2023.0310
18. Grimaldi AF, Chen W, Nordstrom KE. Adherence and persistence in chronic weight management. *Clin Nutr Rep*. 2023;20(1):44–61. doi:10.8180/cnr.2023.0044
19. Northvale Therapeutics. ZENITH-3: a randomised, double-blind, placebo-controlled trial of bexanotide in adults with obesity. Clinical Trial Summary Report. 2024;NVT-ZEN3-001.
20. Aleixo CM, Haugen SB, Prakash RN. Primary endpoint results from the ZENITH-3 programme. *J Metab Med*. 2024;51(2):133–152. doi:10.4821/jmm.2024.0133
21. Steinmetz JL, Cabrera OD. Consensus guidance on staging and treatment escalation in obesity care. *Int Rev Endocrinol*. 2024;56(1):12–39. doi:10.3310/ire.2024.0012
22. Lindqvist H, Barnaby TR. Shared decision-making frameworks in metabolic care. *Clin Pract Advis*. 2023;22(8):702–719. doi:10.7788/cpa.2023.0702
23. Osei-Bonsu K, Falk AM. Long-term follow-up standards for weight management programmes. *Ann Obes Sci*. 2024;16(1):55–73. doi:10.5507/aos.2024.0055
24. Northvale Therapeutics Medical Affairs. Internal evidence summary: obesity care pathway. 2024;NVT-MA-0117.